Across-the-board extraction

Fragment 3 — 2026-05-07

This fragment is a deliberate stress test. Pullum's handout shows how a SLASH-free, logic-on-trees treatment can describe ordinary unbounded dependencies, then flags Across-the-Board extraction as the hard residue where Gazdar's GPSG analysis is especially elegant.

1. Phenomenon

Across-the-Board extraction allows one preposed phrase to correspond to gaps inside each coordinate:

(1) *Those, I either sent back or paid for.*

(2) **Those, I either paid the bill or sent back.*

(3) **Those, I either sent back or paid the bill.*

The good case has matching gaps in both coordinates. The bad cases have only one gap: one coordinate is independently compatible with the extracted NP, but the other is not. The fragment asks whether this can be stated directly over trees, without reintroducing a SLASH-valued category.

2. Vocabulary

Reuse Pullum's vacancy and bindability machinery, but add coordination relations:

(4) $\text{Coord}(c)$: c is a coordinate construction.
$\text{Conjunct}(c, k)$: k is a conjunct daughter of coordinate construction c.
$\text{ParallelGap}(z_1, z_2)$: vacancies z_1 and z_2 are syntactically parallel.

The last predicate is a deliberate pressure point. For NP extraction, it might mean that both vacancies are complements of comparable lexical heads, or that both are linked to the same extracted prenucleus NP by the same syntactic relation. It is not a semantic or thematic role predicate. If that extra condition cannot be reduced to syntactic relations such as function-labelled dominance plus a shared prenucleus relation, the fragment has located a real bridge problem rather than solved it.

3. Specification

Use the same basic vacancy definitions:

(5) $\text{Vacancy}(x) \equiv_{\text{def}} \text{Phrasal}(x) \wedge \neg(\exists y)[x \lhd y]$

(6) $\text{NPVacancy}(x) \equiv_{\text{def}} \text{Vacancy}(x) \wedge \text{NP}(x)$

Assume the Pullum-style definition of $\text{NPBindable}(r)$: r roots a bridge-linked dominance chain containing an NP vacancy. Then define a conjunct as NP-binding if it has such a root:

(7) $\text{ConjNPBindable}(k) \equiv_{\text{def}} (\exists r)[k \lhd^* r \wedge \text{NPBindable}(r)]$.
conjunct k contains an NP-bindable domain

The weak ATB condition requires every conjunct to contain a bindable NP gap:

(8) $\text{ATB}(c) \equiv_{\text{def}} \text{Coord}(c) \wedge (\forall k)[\text{Conjunct}(c, k) \Rightarrow \text{ConjNPBindable}(k)]$.
all conjuncts contain an NP-bindable domain

That is not quite enough. It rules out the one-gap cases, but it does not yet guarantee that the gaps are parallel in the right way. A stronger condition quantifies over the actual vacancies:

(9)
$$\begin{aligned}\text{ATB}^+(c) \equiv_{\text{def}}\ & \text{Coord}(c) \wedge (\forall k)[\text{Conjunct}(c, k) \Rightarrow (\exists z)[k \lhd^* z \wedge \text{NPVacancy}(z)]] \\ & \wedge (\forall k_1)(\forall k_2)(\forall z_1)(\forall z_2) \\ & \quad [\text{Conjunct}(c, k_1) \wedge \text{Conjunct}(c, k_2) \wedge k_1 \lhd^* z_1 \wedge k_2 \lhd^* z_2 \wedge \\ & \quad\quad \text{NPVacancy}(z_1) \wedge \text{NPVacancy}(z_2) \Rightarrow \text{ParallelGap}(z_1, z_2)].\end{aligned}$$
all conjuncts contain NP vacancies, and those vacancies are syntactically parallel

4. Worked instance

In *Those, I either sent back or paid for*, the coordinate VP contains two conjuncts. The first has an NP vacancy as complement of the particle-verb complex *sent back*; the second has an NP vacancy as complement of *for*. Each conjunct therefore contains an NP-bindable domain, and the coordinate satisfies the weak ATB condition. If the two missing NPs are both linked to the same prenucleus NP *those*, they also satisfy the intended ParallelGap condition. This is a syntactic bridge condition, not a claim that the two gaps have the same semantic role.

In *Those, I either paid the bill or sent back*, the first conjunct has no NP vacancy corresponding to *those*. It therefore fails ATB(c). The mirror-image example fails for the same reason in the second conjunct.

5. Open questions

(a) ParallelGap is doing real work. If it cannot be reduced to ordinary function-labelled dominance plus the shared prenucleus relation, the fragment has merely hidden a SLASH-like dependency under a new name.

(b) Semantic or thematic role matching is deliberately absent. If ATB requires such a condition, it needs a separate interface predicate with its own warrant.

(c) The formula needs a sharper account of coordination structure: binary vs. flat coordination, markers such as *either* and *or*, and possible non-constituent coordination.

(d) The weak condition may overgenerate cases with accidental gaps that are not jointly licensed by the same filler.

(e) This should be compared directly with Gazdar's 1981 treatment before being treated as a successful replacement.